

REVIEW

Phosphoproteomics in Vascular Biology and Disease: Illuminating Signaling in the Vessel

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ABSTRACT: Phosphoregulatory events underlie vascular responses to environmental and pathological stimuli, regulate hemostasis and thrombosis, and drive vascular remodeling in health and disease. Consequently, defining phosphorylation-dependent signaling networks in vascular physiology and pathology has enabled biomarker discovery and informed therapeutic interventions. Advances in mass spectrometry-based phosphoproteomics have enabled accurate and high-resolution mapping of dynamic phosphorylation events at the systems level, providing comprehensive mechanistic insights into vascular signaling and pathology. In this review, we contextualize phosphorylation in the vascular niche and summarize the current state of phosphoproteomics in vascular research, highlighting experimental design considerations, technological advances, quantification strategies, and data analysis approaches to uncover biological insights from large-scale phosphoproteomic data sets. Finally, we discuss recent discoveries in vascular signaling and disease, along with current challenges and emerging directions for applying phosphoproteomics to critical questions in vascular biology.

Key Words: biology ■ data analysis ■ hemostasis ■ phosphorylation ■ vascular remodeling

Vascular biology is governed by coordinated signaling networks among endothelial cells (ECs), vascular smooth muscle cells (VSMCs), platelets, fibroblasts, and immune cells (eg, vascular macrophages). Phosphorylation, in addition to other posttranslational modifications (PTMs), such as acetylation, ubiquitination, and redox modifications, plays a central role in regulating vascular signaling networks. Protein phosphorylation has long been considered one of the most prevalent and functionally significant PTMs in cellular signaling,¹ acting as a reversible molecular switch that controls kinase-phosphatase activity and regulates protein-protein interactions, localization, metabolic programming, and more.² Vascular physiology requires rapid, context-specific phospho-responses, such as shear stress sensing, barrier modulation, thrombosis, vasoconstriction, and vascular repair and remodeling, making the capture of such transient events crucial to understanding physiology and disease in vascular niches.

Mass spectrometry (MS)-based phosphoproteomics, herein referred to as phosphoproteomics, is a powerful tool that captures dynamic signaling events and provides an unbiased global quantitative readout of signaling networks far beyond what single-protein

measurements or targeted kinase assays can provide. Over the past 2 decades, advances in phosphoproteomics have enabled high-throughput quantitative profiling of tens of thousands of phosphorylation events across diverse biological conditions.^{3,4} Innovative developments in sample processing, phosphopeptide enrichment strategies, liquid chromatography (LC) separations, and advanced MS instrumentation and acquisition methods have increased the capability for systems-level mapping of phosphorylation with greatly improved depth, sensitivity, reproducibility, and throughput.^{3,4} The integration of phosphoproteomics data with computational analysis has further allowed researchers to move beyond descriptive catalogs of phosphosites toward pathway-level interpretations and the identification of actionable signaling nodes,^{5–7} making phosphoproteomics an indispensable tool for decoding vascular signaling. At the same time, important technical challenges remain in applying phosphoproteomics to vascular research, including cell and tissue heterogeneity, the low abundance and low stoichiometry of many regulatory phosphorylation events, and the limited quantity and variable quality of clinically relevant samples.^{8,9}

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Arterioscler Thromb Vasc Biol is available at www.ahajournals.org/journal/atvb

Nonstandard Abbreviations and Acronyms

APC	activated protein C
DIA	data-independent acquisition
EC	endothelial cell
IMAC	immobilized metal affinity chromatography
LC	liquid chromatography
m/z	mass-to-charge ratio
MS	mass spectrometry
MS/MS	tandem mass spectrometry
PDK	pyruvate dehydrogenase kinase
PTM	posttranslational modification
VEGF	vascular endothelial growth factor
VSMC	vascular smooth muscle cell

In this review, we provide a framework for applying phosphoproteomics to vascular biology research. We first provide an overview of how key phosphoregulatory events promote cell function in the vascular niche. We then summarize current workflows for phosphoproteomics and highlight recent applications that have expanded our knowledge of vascular signaling, disease, and translational applications. We further provide insight into the application of advanced phosphoproteomics to enhance understanding of vascular biology in the future.

PHOSPHORYLATION IN THE VASCULAR LANDSCAPE: A BRIEF OVERVIEW

The vessel wall is a highly organized, multicellular structure composed of ECs, VSMCs, fibrotic cells, resident immune cells, and an extracellular membrane. Capillaries consist of a single endothelial layer supported by pericytes, whereas larger veins and arteries contain multiple basement membranes and concentric layers of VSMCs that regulate vascular tone and mechanical integrity. In addition to resident vascular cells, circulating platelets, progenitor cells, and immune cells are dynamically recruited to the vessel wall, where they modulate inflammation, hemostasis, and vascular remodeling. Together, these cellular components form the vascular niche, a signaling-rich microenvironment in which mechanical, chemical, and receptor-mediated cues regulate vasomotion, including vascular contraction and relaxation, as well as cellular growth responses such as hypertrophy and hyperplasia, in addition to changes in cellular migration, activation, differentiation, and senescence. Across this landscape, reversible protein phosphorylation serves as a central integrative mechanism that translates diverse extracellular cues into coordinated cellular responses.

ECs maintain vascular homeostasis and orchestrate remodeling and angiogenesis through tightly controlled

Highlights

- Phosphoproteomics is a powerful approach for mapping dynamic signaling within the vascular niche that cannot be captured by transcriptomic or proteomic profiling alone.
- Technological advances now enable the identification of thousands of phosphosites from samples ranging from low-input (<10 µg) samples to bulk tissue analyses and large-scale clinical cohorts.
- Computational tools for downstream analysis and functional interpretation continue to evolve alongside increasingly large and complex phosphoproteomic data sets.
- Phosphoproteomics has defined physiological and disease-associated signaling networks in endothelial cells, macrophages, and vascular smooth muscle cells; however, emerging advances in instrumentation, low-input, and spatial technologies offer substantial untapped opportunities for vascular discovery.

phosphorylation-dependent signaling networks. The balance between vasodilatory and vasoconstrictive mediators, regulation of endothelial barrier function, and coordination of platelet interactions and coagulation are all governed by dynamic phosphorylation events that enable rapid and context-dependent responses to metabolic, inflammatory, and hemodynamic stimuli.¹⁰ Mechanosensory complexes at endothelial junctions further couple shear stress and stretch to intracellular phosphorylation cascades that control cell alignment, junctional stability, and adaptive remodeling.¹¹ During angiogenesis, phosphorylation-dependent signaling downstream of growth factor receptors and adhesion complexes directs endothelial survival, sprouting, migration, and vessel stabilization, ensuring spatial and temporal coordination of new vessel growth.¹²

VSMCs are the principal contractile cells of the arterial wall and play a key role in regulating vessel tone, blood pressure, and regional blood flow. Phosphorylation-dependent control of calcium handling, cytoskeletal organization, and actomyosin interactions enables precise modulation of contractility.¹³ Beyond contraction, VSMCs actively synthesize and remodel the extracellular matrix, with phosphorylation-driven signaling pathways governing matrix production, organization, and the mechanical properties of the vessel wall.¹⁴ In response to biochemical cues such as growth factors, inflammatory mediators, hypoxia, or biomechanical stress, these signaling networks drive transitions from a quiescent, contractile phenotype toward proliferative, migratory, inflammatory, and matrix-remodeling states.¹³ Dysregulation of these phosphorylation-dependent programs underlies pathological vascular remodeling observed in hypertension, atherosclerosis, restenosis, and vascular calcification.¹³

Immune cells, particularly macrophages, represent another critical phosphorylation-regulated component of the vascular niche. Macrophages reside within the vessel wall and are recruited from circulating monocytes during injury and disease, where they integrate environmental signals to coordinate inflammation, tissue repair, and remodeling.¹⁵ Phosphorylation-dependent signaling pathways regulate macrophage activation, polarization, lipid handling, cytokine production, and interactions with vascular cells.¹⁵ In atherosclerosis, these processes govern foam cell formation, extracellular matrix remodeling, efferocytosis, and plaque stability.¹⁶ The high plasticity of macrophage phenotypes reflects the dynamic and context-specific nature of phosphorylation signaling within the vascular environment, with important consequences for disease progression and resolution.

Together, these examples demonstrate that phosphorylation functions as a dynamic regulatory mechanism in the vascular niche, coordinating signal integration, amplification, and feedback across cell types and spatial domains. Despite extensive functional characterization of key kinases and phosphatases that regulate vascular function, important gaps remain in defining the specific functional phosphosites that encode vascular cell states and determining how these sites are rewired in disease. Emerging phosphoproteomic approaches now provide the resolution and scale needed to map these events across vascular tissues and cell types (Figure 1). By extending these approaches to signaling in human primary cell models and tissue biopsies, phosphoproteomics has informed disease mechanisms, guided therapeutic target discovery, and nominated biomarkers to monitor diseases, such as aortic aneurysm, pulmonary hypertension, and atherosclerosis. In the following sections, we summarize the current state of phosphoproteomics technologies and highlight recent applications in vascular research.

CURRENT PHOSPHOPROTEOMIC WORKFLOWS

The goal of phosphoproteomics is to conduct quantitative, in-depth profiling of the phosphoproteome, often in parallel with the global proteome from the same set of samples (Figure 2A and 2B), recognizing that changes in protein abundance and phosphorylation are tightly interconnected. The following sections outline a typical phosphoproteomic pipeline, including experimental design, protein extraction and digestion, phosphopeptide enrichment, LC–tandem MS (MS/MS) analyses, and subsequent statistical, bioinformatics, and structural analysis steps (Figure 2). For more detailed reviews of technological advances in phosphoproteomics, please refer to several recent reviews.^{4,17}

Experimental Design and Sample Preparation

In vascular biology, rigorous experimental design is essential for generating biologically meaningful and reproducible data. Studies should be anchored in a clear hypothesis-driven aim, such as determining the effects of acute or chronic shear stress, defining mechanisms of growth factor signaling, or contrasting vascular phenotypes in health and disease. Figure 1 summarizes types of samples and related experimental aims that can be derived in vascular phosphoproteomics.

Sample preparation begins with careful preservation of phosphorylation states through rapid lysis under phosphatase-inhibiting conditions, as well as minimization of artifactual signaling induced by tissue handling, ischemia, or platelet activation. Protein lysates are then digested into peptides (Figure 2A), typically with trypsin alone or in combination with Lys-C, and enriched for phosphopeptides to increase phosphosite coverage. Sample preparation workflows are increasingly compatible with automation (eg, SP3

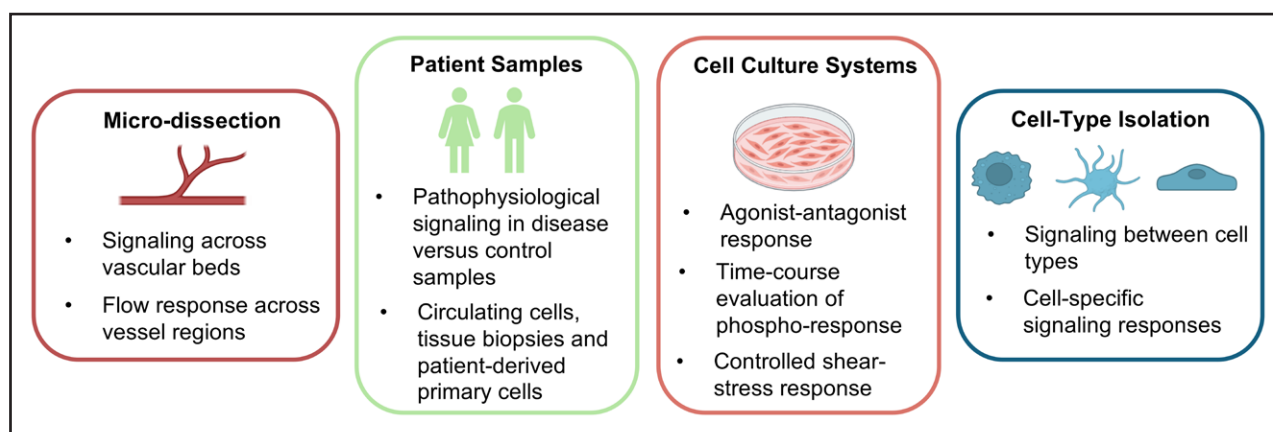


Figure 1. Vascular phosphoproteomics sample types and applications.

Human patient samples, isolated vascular cell types and microdissected vessel segments, and controlled cell culture systems are integrated to study disease-associated signaling, cell-specific phosphorylation responses, tissue- and region-specific signaling across vascular beds, and dynamic agonist-dependent and shear stress-dependent phosphoproteome changes.

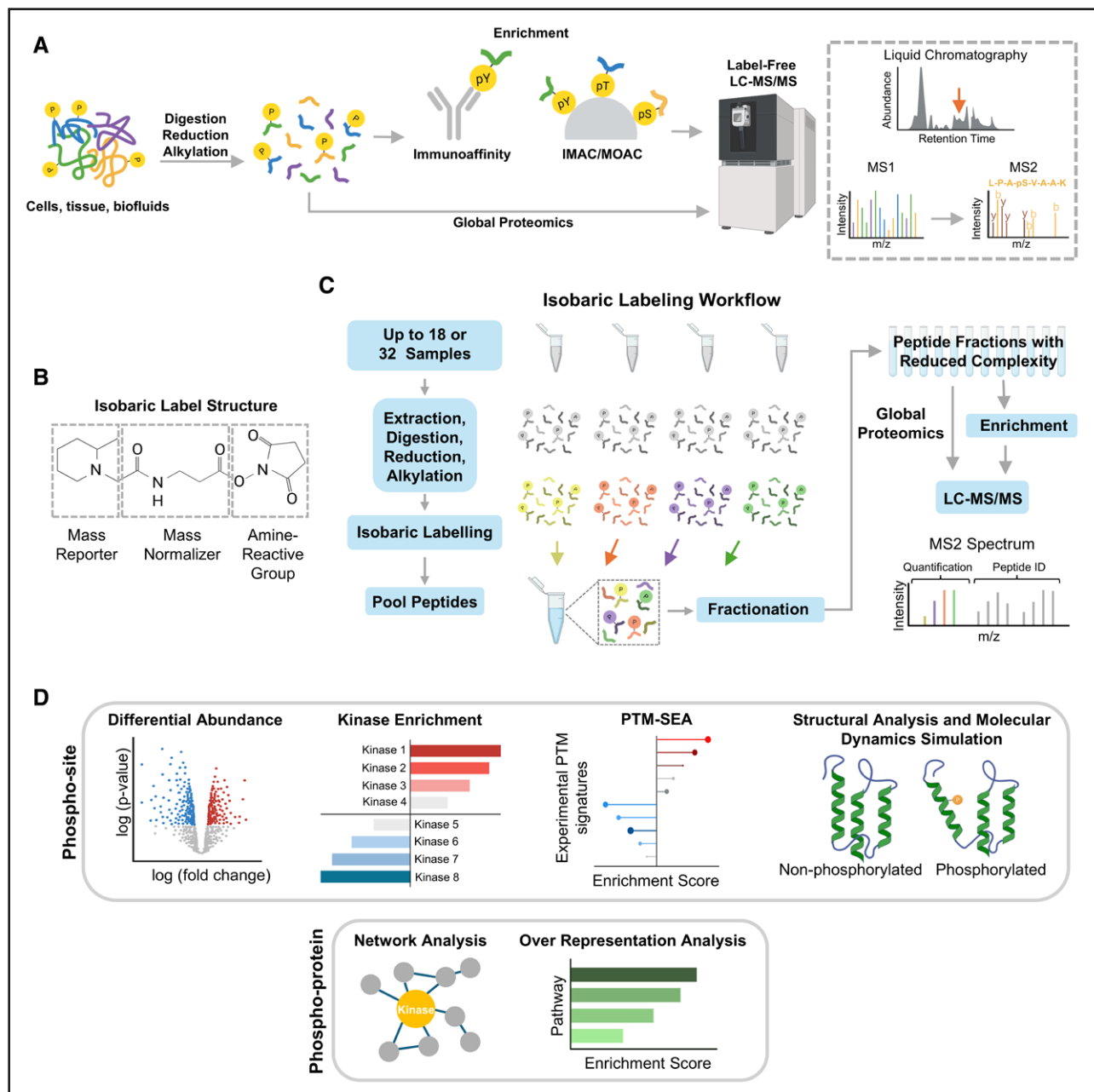


Figure 2. Overview of phosphoproteomic workflows.

A, Sample processing starts with protein digestion, alkylation, and reduction. For phosphoproteomic analysis, digested peptides are enriched for phosphopeptides. Common enrichment strategies include immunoaffinity enrichment for phosphotyrosine (pY) peptides or metal affinity-based enrichment, which enriches phosphotyrosine, phosphoserine (pS), and phosphothreonine (pT) peptides. In liquid chromatography–tandem mass spectrometry (LC-MS/MS) analysis, the precursor mass-to-charge ratio (m/z) is determined in the first mass spectrometry scan (MS1), whereas peptide sequences are determined in the second MS scan (MS2). **B** and **C**, Isobaric labeling uses amine-reactive tags with a mass reporter (**B**) for each biological sample. Samples can then be pooled, enriched, and analyzed (**C**). Reporter ion intensities enable relative quantification during MS2 analysis. **D**, Downstream bioinformatics of phosphosite-specific quantitative data typically includes differential abundance, kinase enrichment, posttranslational modification (PTM)–set enrichment analysis (SEA), and structural analysis. Phosphoprotein-level analyses can further include protein-protein interaction networks and pathway enrichment. IMAC indicates immobilized metal affinity chromatography; and MOAC, metal oxide affinity chromatography.

paramagnetic beads) and liquid-handling robotics,^{18,19} facilitating robust, reproducible phosphoproteomic measurements capable of capturing dynamic vascular signaling events.

Phosphopeptide Enrichment

Phosphorylation, like many other PTMs, is present at low stoichiometry, necessitating enrichment strategies

to ensure broad phosphoprotein coverage (Figure 2A). The 2 most commonly applied affinity-based methods for phosphopeptide enrichment include immobilized metal affinity chromatography (IMAC)²⁰ and metal oxide affinity chromatography.²¹ Both IMAC and metal oxide affinity chromatography enrich phosphopeptides using chelated metal ions (eg, Fe³⁺ or Ga³⁺) or TiO₂, respectively, to exploit the strong affinity of negatively charged phosphate groups for metal ions or metal oxide surfaces. To increase phosphopeptide coverage, LC fractionation, especially high-pH reversed-phase LC, is often incorporated either before or after phosphopeptide enrichment,^{3,22} separating single samples into 6 to 12 phosphopeptide fractions for LC-MS/MS analyses. Recent innovations include automated, high-throughput workflows that leverage robotic liquid handling (King-Fisher and Bravo/AssayMAP), incorporating either magnetic IMAC beads²³ or cartridge-based IMAC²⁴ to reduce sample variation and decrease processing time.

Most established workflows are optimized for bulk samples (eg, >100 µg protein per sample) and often lack the sensitivity or coverage required for low-input samples (<10 µg protein). Recent developments have improved coverage for small sample volumes through techniques such as one-pot workflows,²⁵ plate-based µPhos,²⁶ and tandem tip-based IMAC,²⁷ enabling analysis of low-µg to even low-ng material from scarce clinical or microdissected samples. Approaches that isolate defined cellular or spatial compartments, such as cell sorting, laser-capture microdissection, or region-specific phosphoproteomic profiling, further facilitate analysis of low-input material and offer cell-specific or spatial signaling information.²⁸ These strategies are particularly relevant for studies of small resistance vessels, discrete plaque regions, or limited human biopsy specimens.⁹ Another challenge in phosphoproteomics is the limited coverage of tyrosine phosphorylation, which is particularly important in vascular biology, where receptor and nonreceptor tyrosine kinases regulate numerous signaling events.²⁹ Because tyrosine phosphorylation is both functionally significant and relatively low in abundance, immunoaffinity enrichment using tyrosine phosphorylation-specific antibodies immobilized on magnetic beads is commonly used to enhance coverage and enable detailed analysis of tyrosine phosphorylation-mediated signaling in vascular samples (Figure 2A).^{30,31} Overall, advances in magnetic bead-based affinity reagents, automated workflows, and microscale techniques have substantially increased the reproducibility and scalability of phosphopeptide enrichment, enabling broad applications across various sample types, including microscale samples for interrogating critical biological questions in the vascular niche.

Quantification Strategies

Careful sample preparation and enhanced enrichment strategies lay the foundation for robust detection and

quantification of phosphopeptides. Most quantification strategies are applicable to both global proteomics and phosphoproteomics, but each approach offers distinct advantages depending on the experimental design.

The most straightforward approach is label-free quantification, in which samples are run independently and compared in downstream analyses. This approach requires minimal sample preparation but often increases instrument runtime and can introduce run-to-run variability. However, increasing robustness of LC-MS instrumentation and data-independent acquisition (DIA), discussed in the next section, have made label-free quantification increasingly robust and commonly applied.³²

Alternatively, isobaric labeling strategies, such as tandem mass tag³³ labeling, provide peptide-level barcoding of up to 18 or 32 samples, which can then be combined for a single LC-MS/MS analysis (Figure 2B and 2C). LC fractionation (eg, high-pH reversed-phase LC) is often incorporated into tandem mass tag workflows, to further enhance peptide coverage.^{32,34} Multiplex labeling strategies allow simultaneous measurements across many conditions, time points, or patient samples in a single LC-MS/MS experiment, minimizing missing data and increasing throughput, reproducibility, and within-plex comparability.

Instrumentation and Data Acquisition

Regardless of the quantification strategy, the quality and reproducibility of phosphoproteomic data depend on the underlying instrumentation and data acquisition. Efficient peptide separation, sensitive ionization, and precise mass analysis are essential to detect low-abundance phosphopeptides and accurately measure their relative levels across samples. To achieve this, LC-MS/MS workflows typically couple a nanoLC system with a high-resolution, accurate-mass instrument (eg, Orbitrap or time-of-flight platform). NanoLC-MS/MS uses low-nanoliter-per-minute flow rates and narrow-bore columns to concentrate phosphopeptides in small elution volumes, boosting ionization efficiency and detection sensitivity. Once phosphopeptides are delivered and ionized, their identification and quantification rely on MS/MS.

MS/MS measures both peptide abundance and sequence identity. During the first MS scan, the mass-to-charge ratio (*m/z*) of each peptide, or precursor ion, is measured. Selected precursor ions are then fragmented and analyzed in a second MS scan. The resulting second MS scan spectra are computationally analyzed to identify peptide sequences and specific phosphosites using database-searching algorithms, such as MSFragger40³⁵ (Figure 2A). In conventional data-dependent acquisition mode, each first MS scan is followed by multiple second MS scan scans of the top *N* most abundant precursor ions for peptide sequencing. Although data-dependent acquisition is widely adopted, stochastic sampling biases

higher-abundance peptides and may miss lower-abundance or coeluting phosphopeptides. By contrast, in DIA mode, each first MS scan is followed by systematic fragmentation across a series of predefined m/z windows spanning the full mass range (eg, 400–1200 m/z). In each m/z window (eg, 2–4 m/z), all coeluting precursor ions are fragmented and analyzed, generating composite second MS scan spectra from multiple coisolated peptides. This requires advanced computational algorithms for data processing, identification, and quantification.³⁶ Although current implementations are generally not used with isobaric tags,³⁷ adoption of DIA has become increasingly popular due to markedly improved the reproducibility and coverage of the phosphoproteome, throughput of analyses, and availability of instrumentation and computational tools.⁴

The selection of an appropriate workflow, from quantification strategy to data acquisition, depends on practical study design considerations, including clinical cohort size, available sample amounts, and intended analytical goals, such as the detection of low-abundance phosphosites or accurate quantification of dynamic signaling events. The Table summarizes common integrated workflows with the technologies discussed above, including practical guidance on use cases and relative cost. It should also be noted that recent instrumentation advances,

including the Orbitrap Astral and trapped ion mobility spectrometry–time-of-flight mass spectrometers, have further enhanced sensitivity and speed for peptide identification. For example, a recent study reported that the Astral coupled with DIA enables mapping of $\approx 30\,000$ unique phosphorylation sites within 30 minutes of data acquisition.³⁸ Continued innovation in MS technologies is expected to further expand the depth, speed, and applicability of phosphoproteomics.

Data Integration and Network Inference

Advances in sample preparation, phosphopeptide enrichment, quantification strategies, and high-throughput MS instrumentation now generate data sets containing tens of thousands of phosphopeptide abundance measurements and site-specific phosphorylation events. Public proteomics repositories, such as PRIDE and MassIVE, within the ProteomeXchange Consortium,³⁹ also contain previously acquired data sets that can be reanalyzed under specific biological lenses, representing a largely underutilized resource for hypothesis-generating research. Although vascular-specific data sets remain relatively limited, these resources nevertheless provide valuable opportunities for secondary analysis and cross-study comparison.

Table. Comparative Summary of Common Phosphoproteomics Workflows

Integrated workflow*	Sample type	Features	Limitations	Relative cost
Label-free DDA	Bulk samples (>100 μ g proteins)	- Moderate phosphoproteome coverage - Suitable for discovery studies	- Lower run-to-run reproducibility with missing data - Biased toward high abundance phosphosites	\$
Label-free DIA	Bulk samples (>100 μ g proteins)	- Moderate to high phosphoproteome coverage - Excellent run-to-run reproducibility - Ideal for large-scale or longitudinal studies	- Requires advanced instrumentation - Needs specialized software and complex data analysis	\$\$
One-POT, μ Phos	Low-input samples (<10 μ g), including microdissected or near-single-cell material	- Streamlined workflow - Minimal sample loss - Ideal for spatial or near-single-cell resolution	- Reduced depth of coverage compared with bulk workflows - Often requires automation of low-volume steps	\$\$
TMT-DDA	Bulk samples (>100 μ g proteins), up to 18-plex per experiment	- Precise relative quantification within a single batch - Deepest coverage - Ideal for time-course studies or small clinical cohorts	- Complex sample preparation - Expensive reagents - Often requires fractionation for high proteome coverage	\$\$
Stable isotope labeling by amino acids in cell culture–DDA	Bulk samples (>100 μ g proteins) from live cell culture, up to 3-plex per experiment	- Highly accurate relative and dynamic quantification - Ideal for pulse-chase and turnover experiments	- Requires active protein synthesis for labeling incorporation - Limited applicability to nondividing cells - Limited multiplexing capacity	\$\$
Targeted MS (SRM/PRM)	Bulk samples (>100 μ g proteins)	- High sensitivity, specificity, and precision of quantification - High analytical throughput with limited targets - Suitable for validation studies	- Assay development can be time consuming - Requires prior knowledge of targets	\$†

DDA indicates data-dependent acquisition; DIA, data-independent acquisition; IMAC, immobilized metal affinity chromatography; LC-MS/MS liquid chromatography–tandem mass spectrometry; MOAC, metal oxide affinity chromatography; PRM, parallel reaction monitoring; SRM, selected reaction monitoring; and TMT, tandem mass tag.

*All workflows involve digestion, IMAC/MOAC, and LC-MS/MS as outlined in Figure 2A.

†

Relative cost per run is low, but the overall cost depends on assay development and the use of isotope-labeled standards.

To assign biological meaning to these complex data sets, integrative computational and pathway analysis approaches are used to organize, interpret, and contextualize phosphoproteomic measurements, as illustrated in Figure 2D. Inferring kinase activity is a widely used strategy for interpreting phosphoproteomic data sets and for linking phosphorylation changes to upstream signaling mechanisms, including those relevant to disease. Widely used methods for kinase inference include Kinase-Substrate Enrichment Analysis³⁸ and PTM Signature Enrichment Analysis,⁴⁰ which estimate upstream kinase activities based on changes across sets of known substrates (phosphopeptides).⁴⁰ Although powerful, these approaches rely on existing kinase-substrate annotations and are therefore restricted to characterized phosphosites. Motif-based prediction provides a complementary strategy for inferring upstream kinases at uncharacterized sites based on local sequence context. Integrative tools such as NetworKIN⁴¹ combine motif scoring with predicted or known protein-protein interactions to more accurately infer kinase-substrate relationships and pathway activity, enabling the prediction of signaling events for kinases lacking direct experimental annotation.

Beyond kinase-centric analyses, pathway-focused approaches can help interpret phosphorylation changes for proteins or sites without known functional significance. Overrepresentation analysis or rank-based enrichment methods, analogous to gene set enrichment analysis, map regulated phosphosites or parent proteins onto curated pathway databases (eg, KEGG, Reactome, Gene Ontology) to identify differentially phosphorylated pathways or biological processes.⁴² Data-driven signaling network inference can further reveal potential interactions and signaling relationships, although current methods remain constrained by incomplete kinase-substrate annotations and context specificity.⁴³ Finally, with structural information from resources such as AlphaFold and the Protein Data Bank, the impact of site-specific phosphorylation on protein function can now be systematically analyzed.^{44,45} Integration of kinase inference, pathway enrichment, network analysis, and structural context thus represents a powerful strategy for extracting biologically meaningful insights toward understanding phosphorylation in vascular biology.

From Phosphosites to Functional and Translational Insight

Often, the goal of phosphoproteomic studies is not merely to catalog phosphorylation events, but to identify key signaling nodes that drive disease processes, cell-state transitions, and cellular responses to perturbation. Despite substantial advances in computational tools, downstream phosphoproteomic analyses continue to provide only a partial view of signaling network activity. As such, their outputs are best interpreted as

hypothesis-generating rather than fully explanatory, and the integration of orthogonal experimental strategies remains essential for robust biological inference.

At the level of kinase-enrichment analysis, inferred kinase activity can be corroborated by assessing phosphorylation at known activating sites using orthogonal methods such as Western blotting, which provides independent confirmation of predicted signaling changes. However, extending such inference from kinase-level activity to the functional interpretation of individual phosphosites remains considerably more challenging. This reflects the fact that phosphorylation events arising from distinct upstream pathways are often codetected under the same experimental perturbations, and not all identified sites are functionally relevant.

Prioritization of candidate functional phosphorylation events from large data sets can therefore benefit from multiple lines of evidence, including known disease relevance of the protein or site, evolutionary conservation of the modified residue across species, and localization within catalytic domains or other structurally and functionally critical regions. In addition, structural modeling approaches mentioned in the previous section can be used to assess whether site-specific phosphorylation is likely to induce conformational changes that may alter protein function.

Once candidate sites have been prioritized, experimental validation is required. When suitable reagents are available, antibody-based detection of site-specific phosphorylation (eg, phospho-specific Western blotting) remains a widely used and accessible approach. However, antibody development for novel phosphosites is often costly and may not be feasible when multiple candidate sites require evaluation. In such cases, targeted MS-based approaches, such as selected reaction monitoring or parallel reaction monitoring, provide a higher-throughput and highly quantitative strategy for phosphosite validation. Targeted MS enables precise and reproducible quantification of predefined peptides with phosphorylation sites of interest identified in discovery phosphoproteomics. Finally, following biochemical validation, functional assessment using site-directed mutagenesis, such as phosphomimetic and nonphosphorylatable mutants, can further define the role of specific phosphorylation events in regulating protein function and cellular behavior.

By combining computational prioritization, structural analysis, biochemical validation, and functional assessment, this tiered framework provides a tractable path from global phosphoproteomic data sets toward mechanistic insight into physiology and disease-relevant signaling.

PHOSPHOPROTEOMICS IN VASCULAR BIOLOGY

Advances in phosphoproteomic workflows and data analysis tools now enable high-resolution mapping of

signaling networks across diverse vascular tissues and cell types. These approaches are now being applied to interrogate signaling events and disease mechanisms with direct translational relevance. In this section, we highlight how phosphoproteomics has been used to study vascular biology, with a focus on studies that have deepened our understanding of vascular niche dynamics and driven translational insights.

Phosphoproteomics has been widely used to investigate signaling in ECs. Early investigations of VEGF (vascular endothelial growth factor)-mediated and angiopoietin-mediated signaling established the importance of phosphorylation events during angiogenesis.⁴⁶ These studies identified both shared angiogenic pathways and distinct signatures associated specifically with vascular permeability or vessel stabilization.⁴⁶ Building on this foundation, a recent phosphoproteomic study mapped the temporal VEGF-A/VEGFR2 signaling cascade in ECs, uncovering 33 previously unreported phosphoproteins.⁴⁷ In parallel, investigations combined secretome profiling with phosphoproteomics to identify novel proangiogenic cardiac-secreted factors, METRL and CREDL2, which drive VEGF-independent angiogenesis in coronary artery ECs.^{6,7} These findings highlight alternative signaling mechanisms that can bypass canonical growth factor pathways under stress conditions. Separately, the cytoskeletal protein β IV-spectrin was shown to suppress angiogenesis through ERK-dependent phosphorylation of DNA methyltransferase 1, revealing a previously unrecognized connection between kinase-scaffold regulation and epigenetic control of vascular growth.⁴⁸ Together, these high-throughput phosphoproteomic studies have expanded our understanding of factors regulating angiogenesis and provide new candidates for functional investigation.

The same approaches have provided key insights into EC inflammatory and hemostatic signaling, revealing temporal phosphorylation events that coordinate vascular responses. Integrative multiomic profiling of cytokine-treated ECs has revealed rapid phosphorylation waves that engage NF- κ B, MAPK, and JAK/STAT signaling.⁴⁹ Although this initial profiling identified a limited number of phosphosites, it demonstrated that early phosphorylation events precede transcriptional changes. These events prime ECs for increased permeability and leukocyte recruitment, providing a dynamic view of signaling kinetics not previously characterized. Similarly, high-depth, time-resolved phosphoproteomics has dissected GPCR signaling in ECs, particularly in response to PAR1 stimulation.^{50,51} These studies revealed that thrombin and APC (activated protein C) generate temporally and functionally distinct phospho-signatures. Thrombin rapidly targets cytoskeletal and junctional proteins to drive early barrier disruption, whereas APC preferentially engages nuclear and RNA-processing modules, promoting delayed barrier stabilization and cytoprotection.⁵² Extending this

approach to a broader panel of GPCR agonists, including histamine and additional PAR ligands, uncovered a conserved phosphorylation backbone centered on cell-cell junctions and actin remodeling, identifying a shared effector layer that integrates diverse vascular signals.⁵¹ Recent analyses of platelet-endothelial communication demonstrate how EC signaling inhibits platelet activation, uncovering novel insights into bidirectional hemostatic-endothelial crosstalk.⁵³ Collectively, these phosphoproteomic investigations provide new insights into the temporal logic of endothelial signaling and reveal conserved molecular mechanisms that may be targeted to fine-tune vascular barrier function.

Beyond chemical cues, phosphoproteomics has clarified how mechanical forces shape endothelial signaling programs. For example, phosphoproteomic analyses have identified BMP9/10 signaling as a key flow-adaptation axis that transduces phosphoprotein signals independently of canonical SMAD pathways and have uncovered additional mechanisms linking mechanical stress to cell cycle regulation in endothelial quiescence.⁵⁴ Optogenetic activation coupled with DIA has further identified AKT1 as a central mediator of shear stress-related and angiogenesis-related signaling, revealing stimulus-specific phosphorylation programs downstream of this kinase.⁵⁵ Collectively, these studies demonstrate that diverse external cues rapidly remodel endothelial phosphoproteomes to regulate junctional integrity and actin dynamics while preserving distinct kinase logic. This phospho-encoding enables ECs to discriminate between permeability-promoting and barrier-stabilizing programs across growth factor, GPCR, inflammatory, and mechanosensory inputs.^{51,52,54}

In addition to ECs, critical phosphoproteomics studies in VSMCs have highlighted converging mechanisms of VSMC plasticity, linking VSMC stimuli to phenotypic switching across vascular beds. In injury-like models, integrated proteome-phosphoproteome analyses revealed coordinated suppression of contractile machinery alongside ERK- and CXCR4-linked phosphorylation programs driving migration.⁵⁶ Time-resolved studies under cyclic stretch further mapped early PKC-family signaling converging on Akt and ROCK1, supporting multiphase mechanotransduction that precedes longer-term cytoskeletal remodeling.⁵⁷ Complementing these phosphosite maps, diabetic stress promotes VSMC calcification by augmenting mTORC2/Rictor-dependent Akt phosphorylation at S473, thereby enhancing Runx2 transactivity.⁵⁸ Conversely, sodium dichloroacetate promotes osteogenic switching through an AKT-independent p38-Runx2 axis, in which p38 phosphorylation enables Runx2 activation and calcification.⁵⁹ These data offer a mechanistic blueprint connecting signaling pathways to phenotypic remodeling.

Finally, macrophage phosphoproteomics provides a systems view of inflammatory signaling in vascular

disease. Early studies demonstrated that TLR stimulation drives rapid, ligand-dependent serine/threonine phosphorylation programs,⁶⁰ with distinct receptor-specific entry routes that progressively converge on shared MAPK, Rho GTPase, FAK/adhesion, and cytoskeletal modules.⁶¹ Receptor localization further shapes these outputs, encoding trafficking and phagocytic responses within the first minutes of signaling.⁶¹ During macrophage state transitions, including fusion and phagocytosis, phosphorylation emerges as a dominant regulatory layer coordinating cytoskeletal remodeling, metabolism, and vesicle trafficking, often independent of transcriptional change.^{62,63} These studies collectively define conserved phospho-signaling axes that link inflammatory sensing to migration, uptake, and tissue remodeling functions central to vascular lesion progression.

Beyond mapping signaling networks, phosphoproteomics has enabled deeper translational insights into disease mechanisms to inform therapeutics and nominate biomarkers. This is exemplified in atherosclerosis, where disturbed flow and lipid stress promote macrophage-driven inflammatory signaling, while VSMC phosphoregulation directs switching between contractile and synthetic phenotypes.^{64,65} Regional phosphoproteomic profiling of native porcine aortic endothelium comparing areas of laminar versus oscillatory flow at vessel curvatures and bifurcations revealed that oscillatory shear stress activates GRK2 and induces phosphorylation of vinculin at S721.²⁸ Notably, phosphorylation of vinculin at S721 was detectable in serum from patients with coronary artery disease, and endothelial overexpression of a phosphorylation-deficient vinculin^{S721A} mutant reduced plaque formation, supporting a causal role of phosphorylation at this residue in atherogenesis. Complementary studies identified AP-1 as a downstream effector in the G protein-coupled receptor kinase 2 signaling cascade, linking disturbed flow to proatherogenic transcriptional programs.⁶⁶ This work highlights phosphoproteomics as a powerful approach to identify flow-sensitive phosphorylation events that both mechanistically drive atherosclerosis and serve as candidate biomarkers of vascular disease.

Aortic aneurysm progression is distinguished by phosphorylation-mediated loss of VSMC contractile identity and compromise of vessel wall integrity, rendering phosphoproteomic analyses well suited to capture disease-relevant signaling alterations. Studies across experimental models and human tissue consistently demonstrate reduced phosphorylation of core contractile proteins alongside activation of stress-responsive signaling programs that drive VSMC dedifferentiation toward a synthetic, migratory phenotype.^{67–71} Dysregulated phosphorylation of mechanosensing proteins, such as integrins and focal adhesion components, further implicates impaired force transduction in aneurysm progression.⁷² Integrated proteomics and phosphoproteomic analyses

of primary VSMCs isolated from control and abdominal aortic aneurysm patient tissue have recently uncovered sustained dysregulation of kinase signaling, as well as previously unrecognized phosphorylation of glycogen synthase.⁶⁷ Together, these findings highlight aberrant kinase-driven phosphorylation as a central cue that accelerates aortic aneurysm pathogenesis.

The translational potential of phosphoproteomics is further illustrated in pulmonary arterial hypertension, where patient-derived samples reveal widespread remodeling of phosphorylation networks governing proliferation, cytoskeletal dynamics, metabolism, inflammation, autophagy, apoptosis, and hypoxia responses.^{73–75} Complementary phosphoproteomic studies further highlighted ERK5-, JAK2-, and PDGFR β -driven proliferation and glycolytic reprogramming through phosphorylation of enzymes such as PDK (pyruvate dehydrogenase kinase) and pyruvate kinase M2.^{76–79} These findings have directly informed preclinical evaluation of PDK and JAK-STAT inhibitors as candidate therapeutic interventions in pulmonary arterial hypertension.^{80,81}

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FUTURE PERSPECTIVES

Phosphoproteomics has been instrumental in defining vascular signaling modules and elucidating disease mechanisms through the analysis of ECs, VSMCs, macrophages, and vascular tissue. We now have unprecedented data sets capturing thousands of phosphosites that provide insight into signaling and disease progression.

With growing appreciation of vascular complexity and signaling diversity, there is a continuing need for higher-resolution, context-specific techniques to capture these processes in both physiological and disease models. Most phosphoproteomic maps are currently generated in cultured cells under short stimulation windows, leaving open questions about how phospho-signatures vary across vascular beds, sex, aging, and disease, and how well in vitro findings generalize to intact vessels. Although spatial proteomics and imaging studies have already revealed discrete endothelial and perivascular subpopulations with distinct signaling programs⁸ and region-specific phosphorylation changes driven by hemodynamic forces are well established,²⁸ there remains a need for higher-resolution, cell type-specific analyses to capture vascular signaling in vivo. Techniques such as laser-capture microdissection and cell-type purification have begun to resolve spatial contexts,^{9,28,82–84} and emerging microscale workflows, including 1-pot DIA18 and tandem tip-based IMAC20, now enable phosphoproteomic profiling of samples as small as 0.5 μ g or tissue voxels of $\approx$ 200 μ m ($\approx$ 100 cells),^{25,27} making it possible to interrogate subpopulations and subcellular compartments at the phosphorylation level. Applying these advances to vascular pathologies, such as region-specific atherosclerotic

plaques, vascular calcification, and aneurysm-prone vessels, provides an opportunity to uncover cell type-specific phosphorylation programs driving disease.

With the growing wealth and complexity of phosphoproteomic data, there is an opportunity to more directly link kinase activity and site-specific phosphorylation to functional outcomes. Many regulated sites remain uncharacterized, and phosphoproteomic changes are frequently disconnected from quantitative phenotypes, such as permeability, mechanotransduction, leukocyte recruitment, migration, phagocytosis, and fate transitions. Building on prior data sets, experimental strategies could leverage phosphosite-informed perturbations, such as selective activation or inhibition of key kinases in specific vascular subpopulations, combined with single-cell phosphoproteomics and microdissected tissue regions to reveal how ECs, VSMCs, and macrophages coordinate signaling in vivo. This approach was elegantly demonstrated using opto-DIA to dissect AKT-dependent phosphorylation programs⁵⁵ and could be expanded to a variety of other kinases or signaling hubs, or combined with cell-specific promoters in animal models to illuminate site-specific regulation in vivo.

Together, these strategies pave the way for mechanistic, spatially resolved, and physiologically relevant phosphoproteomic atlases that can inform therapeutic development and biomarker discovery in vascular disease.

ARTICLE INFORMATION

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Acknowledgments

Figures and the graphic abstract were created with permission using BioRender.

Sources of Funding

This work was partially supported by National Institutes of Health grants R01DK135081(W.-J. Qian), R01HL178662 (M. He), R01DK144096 (M. He), and R21AG075450 (M. He).

Disclosures

None.

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